Weekly Internship Report (16/5/26 - 22/5/26)

WEEK 1

Intern's Name: Atharv Gupta
Internship Week: Week 1 (16/5/26 - 22/5/26)
Company/Organization Name: Red Kross Research Foundation
Mentor's Name: Shivam Mittal

Week Overview:

- Learned the fundamentals of Cybercrime and Ethical Hacking.
- Explored Malware Analysis techniques including Static, Dynamic, and Behavioural Analysis.
- Created a GitHub repository for internship documentation and cybersecurity project tracking.
- Conducted research on Dev Security App Security (Project Discovery).
- Performed research and analysis on Dridex Malware.
- Completed a Steganography Challenge involving hidden data extraction and SHA-256 hash analysis.
- Performed a Cryptography Exercise using Playfair Cipher and Blake2 in CyberChef.
- Participated in discussions related to AI SOC and Network / ZTNA security features.

Learning Outcomes:

- Understood the basics of Cybercrime and Ethical Hacking concepts.
- Learned different malware analysis methodologies and cybersecurity research techniques.
- Improved GitHub documentation and project management skills.
- Learned the difference between hashing and encryption mechanisms.
- Explored steganography concepts and hidden data extraction techniques.
- Gained practical exposure to cryptographic workflows using Playfair Cipher and Blake2 hashing.
- Developed a basic understanding of AI SOC operations and ZTNA security concepts.

Challenges Faced:

- Understanding malware behavior analysis required detailed observation and research.
- Initially faced difficulty differentiating encryption and hashing concepts.
- Analyzing cryptographic workflows and hidden data extraction processes required additional study.

Goals for Next Week:

- Learn Cyber Threat Intelligence concepts and threat monitoring techniques.
- Explore Digital Forensics and Evidence Handling procedures.
- Improve understanding of cyber investigation workflows.
- Continue enhancing cybersecurity research and documentation skills.

Mentor Feedback and Suggestions:

- Continue improving practical cybersecurity skills and analytical abilities.
- Maintain proper documentation of internship activities and learning outcomes.

Additional Notes:

- This week provided practical exposure to cybersecurity research, malware analysis, cryptography, and cybersecurity documentation workflows.

Student Diary (Log) Entry

Week	Task Assigned	Activities Performed	Key Learnings	Additional Remarks
1	Cybersecurity Fundamentals and Research Tasks	Malware analysis, GitHub repository setup, Dridex research, Steganography challenge, Cryptography exercises, AI SOC and ZTNA discussions	Learned malware analysis, hashing vs encryption, cryptographic workflows, GitHub documentation, and cybersecurity research concepts	Improved practical understanding of cybersecurity operations and research methodologies